

Yichen Hu

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Davis, CA, USA

EDUCATION

- **University of California, Davis** Davis, CA, USA
Ph.D. Candidate in Statistics Sep 2023 – Present
Advisor: Dr. Xiukai Ding.
Research interests: high-dimensional statistics, random matrix theory, manifold learning, graph networks, and applications to data science and machine learning.
- **University of California, Davis** Davis, CA, USA
Master of Science in Statistics Sep 2021 – Jun 2023
- **Shanghai University of Finance and Economics** Shanghai, China
Bachelor of Science in Statistics Sep 2016 – Jul 2020

RESEARCH PROJECTS

Note: [†] denotes that authors are listed in alphabetical order.

- Xiukai Ding, **Yichen Hu**, Zhenggang Wang[†] (2025). Two Sample Test for Covariance Matrices in Ultra-High Dimension. *Journal of the American Statistical Association*, 120(552), 2210–2221.
- Xiukai Ding, **Yichen Hu**[†] (2025). On the Edge Eigenvalues of Sparse Random Geometric Graphs. Submitted to *Annals of Applied Probability*. Available at <https://arxiv.org/abs/2509.07372>.
- **Yichen Hu**, Xiukai Ding, Jian Kang (2026). Adaptive Sparsity-Aware Projection-Assisted Spectral Embedding for High-Dimensional Noisy Data with Nonlinear Structures. Submitted.
- Xiukai Ding, **Yichen Hu**, Jiahui Xie[†]. Threshold-free inference for edge eigenvalues of large covariance matrices via multiplier bootstrap. In progress.
- **Yichen Hu**, Qifeng Lin, Xiukai Ding, Rong Ma. Learning common structures from high-dimensional noisy biviews using cross-integral operator. In progress.

AWARDS

- Peter Hall Graduate Student Research Award, Department of Statistics, University of California, Davis, 2026
- IMS Hannan Graduate Student Travel Award, 2026
- Teaching Recognition Award, Department of Statistics, University of California, Davis, 2025–26
- Outstanding Graduate Student Teaching Award, Department of Statistics, University of California, Davis, 2024–25
- Outstanding Graduate Student Teaching Award, Department of Statistics, University of California, Davis, 2023–24
- Outstanding Master's Student Award, Department of Statistics, University of California, Davis, 2021–22

SOFTWARE

- **RMT4DS**: an R package for random matrix tools in data science.
Available at CRAN: <https://cran.r-project.org/package=RMT4DS>
- **UHDtst**: an R package for two-sample covariance matrix testing in ultra-high-dimensional settings.
Available at GitHub: <https://github.com/EthanHuStat/UHDtst>.

TEACHING

- Teaching Assistant, Department of Statistics, University of California, Davis

- STA 206: Statistical Methods for Research I
- STA 141B: Data & Web Technologies for Data Analysis
- STA 135: Multivariate Data Analysis
- STA 106: Applied Statistical Methods: Analysis of Variance
- STA 108: Applied Statistical Methods: Regression Analysis

SERVICE

- Reviewer for *Electronic Journal of Statistics*.
- Reviewer for *Communications in Statistics – Theory and Methods*.